Bhuvann Vinodh Ram

Graduate AI / Machine Learning Engineer

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Stamp 1G: full-time work authorisation in Ireland

PROFILE

MSc AI for Medicine graduate (UCD) who builds and ships machine learning: CNNs, BiLSTMs, VAEs and gradient boosting on imaging, signal, text and genomics data, with SHAP explanations, leakage checks, tests and CI. Deploys models as Streamlit and Gradio apps and in-browser tools that cost nothing to run. Currently a Trust and Safety Associate at Accenture, Dublin.

SKILLS

Languages: Python, SQL, JavaScript, R (basic)

Machine learning: scikit-learn, XGBoost, ensembles, hyperparameter tuning, leakage checks, time-based validation

Deep learning: TensorFlow/Keras, PyTorch; CNNs, BiLSTM, VAEs, U-Net; transfer learning (ResNet50, VGG16, EfficientNet, MobileNet)

Vision and language: OpenCV, OpenCV.js, Detectron2, Tesseract.js; text classification, embeddings, TF-IDF retrieval

Explainability: SHAP, model interpretation

Shipping models: Streamlit, Gradio, GitHub Pages, Docker, GitHub Actions, pytest

EXPERIENCE

Trust and Safety Associate, Accenture · Dublin, Ireland — Mar 2026 – Present

- Evaluate, classify and moderate user-generated content against client platform policies to keep online communities safe and compliant.

Research Intern, University College Dublin · Dublin, Ireland — Apr 2025 – Sep 2025

- Designed and trained a variational autoencoder over RNA-seq, copy-number and methylation data, then clustered the latent space (K-Means, HDBSCAN).
- Explained the results with SHAP, differential expression and STRING networks, and assessed survival differences with Kaplan–Meier and log-rank tests.

Project Intern, Centre for Toxicology and Development, SRIHER · Chennai, India — Jan 2024 – Jun 2024

- Built a video-to-data pipeline for zebrafish behaviour analysis used in drug testing: tracked fish across tanks and layers, computed speed and velocity, and produced charts and reports.

Research Intern, IIITDM Kancheepuram · Chennai, India — Dec 2022 – Feb 2023

- Developed and compared CNN breast cancer classifiers using pre-trained VGG16, ResNet, MobileNet and EfficientNet models.

Intern, Smartail Pvt. Ltd. · Chennai, India — Feb 2022 – Apr 2022

- Containerised web applications with Docker and deployed them through GitLab CI/CD pipelines.
- Worked on image processing with Detectron2, including hyperparameter tuning.

PROJECTS

Tamil Thedal (தமிழ் தேடல்) · JavaScript · OpenCV.js · Tesseract.js · Python ETL — 2026
Live: bhuvannv13.github.io/tamil-thedal · Code: github.com/bhuvannv13/tamil-thedal

- Search engine for Tamil literature (1,549 verses) with sound-alike matching of romanised Tamil and a TF-IDF chat helper that runs fully in the browser.
- Inscription reader: OpenCV.js enhancement (CLAHE, adaptive threshold), letter segmentation and Hu-moment comparison with Tamil-Brahmi, Grantha and Tamil reference letters; Tesseract.js OCR for modern Tamil.

Conflict Risk Monitor · Python · pandas · scikit-learn · GitHub Actions · JavaScript — 2026
Live: bhuvannv13.github.io/conflict-risk-monitor · Code: github.com/bhuvannv13/conflict-risk-monitor

- Estimates the probability of 25+ battle deaths per country in the next three months; gradient boosting chosen on a validation period scored ROC AUC 0.989 and average precision 0.959 on unseen 2022–26 data.
- In-browser chat assistant (TF-IDF retrieval, no paid API) answers questions from the published data.

ECG heartbeat classification · 1D CNN · Keras 2025
Code: github.com/bhuvannv13/ECG_Classification_CNN

- 1D CNN classifying MIT-BIH heartbeats into five classes: 96.01% test accuracy on 21,892 held-out beats, reproduced by an automated GitHub Actions rerun.

Ovarian cancer survival prediction (HGSOC) · XGBoost · SHAP · TCGA 2025
Code: github.com/bhuvannv13/HGSOC-survival

- XGBoost on TCGA clinical and mutation data with SHAP explanations; found and removed follow-up variables that leaked the outcome, then reran the notebook end to end in GitHub Actions: test accuracy 0.78, ROC AUC 0.91.

Emotion classification from tweets · BiLSTM · TensorFlow · Gradio 2025
Code: github.com/bhuvannv13/Twitter-Emotional-Classification

- Dual-input BiLSTM combining word embeddings with sentiment polarity: 90.63% test accuracy, served as a Gradio app.

Brain tumour classification from MRI · CNN · OpenCV · Streamlit · pytest 2025
Code: github.com/bhuvannv13/Brain_Tumour_classification_api

- Four-class CNN (0.80 accuracy, 0.77 macro F1 on 2,258 test images) and ResNet50 transfer learning, deployed as a Streamlit app with unit tests and CI.

EDUCATION

MSc Artificial Intelligence for Medicine and Medical Research, University College Dublin 2024 – 2025
GPA 3.37 / 4.2

BTech Computer Science and Medical Engineering, Sri Ramachandra Institute of Higher Education and Research, Chennai 2020 – 2024
CGPA 3.18 / 4

AWARDS

- UCD Innovation Challenge 2024: Innovation Spark Award
- NASA Space Apps Challenge 2024 participant
- UCD Advantage Award